PROJECT PROPOSAL

SmartPuck

AI-Powered Meeting & Lecture Recorder

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1. Core Concept

SmartPuck is a compact, low-cost hardware device designed for reliable, passive recording of meetings, lectures, and classroom sessions. Roughly the size of a hockey puck, it captures far-field audio and periodic visual snapshots (e.g., whiteboards, slides).

The system operates completely offline during recording. When plugged into a PC via USB, a companion web application automatically processes the session. It uses AI to transcribe the audio, integrates the images for visual context, and generates structured notes, including summaries, key points, and action items.

The project aims to be a non-intrusive, non-subscription alternative to existing cloud-dependent or phone-based recording tools.

Puck
Board

2. Key Features

- **Offline Operation**: Fully self-contained during recording.
- **Dual Capture**: Records high-quality, far-field audio and captures visual context (images) simultaneously to a local microSD card.
- **Long Battery Life**: Designed to support up to 8 hours of continuous recording.
- **AI-Powered Notes**: Local web app uses a speech-to-text service (e.g., Whisper) and a multimodal language model (e.g., Gemini 3/GPT-5) to produce a structured summary and action items.
- **Affordable Hardware**: Target Bill of Materials (BOM) cost is under $50 USD.

3. Solution Overview

3.1 Device (Phase 1 — Recording)

The device operates in two sequential modes: Recording Mode and Transfer Mode.

During Recording Mode, the ESP32-CAM microcontroller runs two parallel capture loops:

- The ICS-43434 I2S MEMS microphone records audio continuously, writing compressed .mp3 chunks to the microSD card.
- The OV5640 5MP camera captures a JPEG snapshot every 2–10 seconds (user-configurable), saving each image alongside the audio.

The onboard IP5306 power management IC handles LiPo battery charging via the USB port, providing up to 8 hours of continuous recording on a 2000 mAh cell.

3.2 Transfer (Phase 2 — USB Connection)

When the user plugs the device into a PC, the CH340X USB bridge exposes the microSD card as a mass storage device. The companion web app — a browser-based single-page application — detects the device, reads the session directory, and pulls the latest audio and image files.

3.3 AI Processing (Phase 3 — Note Generation)

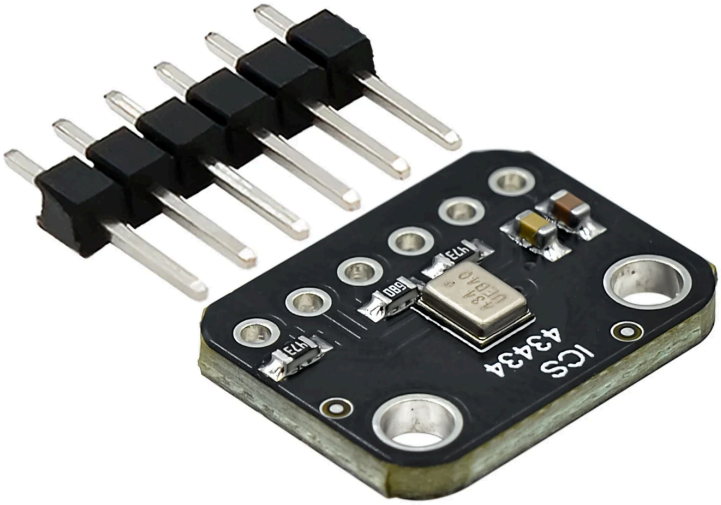
The web app processes the session in two parallel pipelines:

- Audio Pipeline: The .mp3 file is sent to a speech-to-text service (e.g., OpenAI Whisper) to produce a full transcript with timestamps.
- Image Pipeline: Key snapshots are selected based on visual change detection, then passed to a multimodal language model as contextual images.

Both outputs are combined into a single prompt to a large language model, which produces a structured summary: a brief overview, key discussion points, decisions made, and a list of action items with owners.

4. Hardware Specification

Component	Specification
Main Board	ESP32-CAM (classic ESP32, 8MB PSRAM, TF microSD slot, CH340X USB)
Camera	OV5640 5MP — optimised for whiteboard legibility

Component	Specification
Microphone	ICS-43434 I2S MEMS breakout — far-field, low-noise 
Storage	microSD 16–32 GB, Class 10 / U1

Component	Specification
Power	3.7 V LiPo 1000–2000 mAh (JST-PH 2.0) + IP5306 PMIC
Connectivity	USB via CH340X (data + charging)
Enclosure	Custom 3D-printed ABS shell, ~80 mm diameter

5. Software Architecture

5.1 Firmware (ESP32)

- Language: C++ (Arduino / ESP-IDF framework)
- Audio: I2S DMA driver → ring buffer → SD write in 5-second chunks
- Camera: OV5640 driver with auto-exposure → JPEG encode → SD write
- File system: FAT32 on microSD; sessions stored as /sessions/YYYYMMDD_HHMMSS/

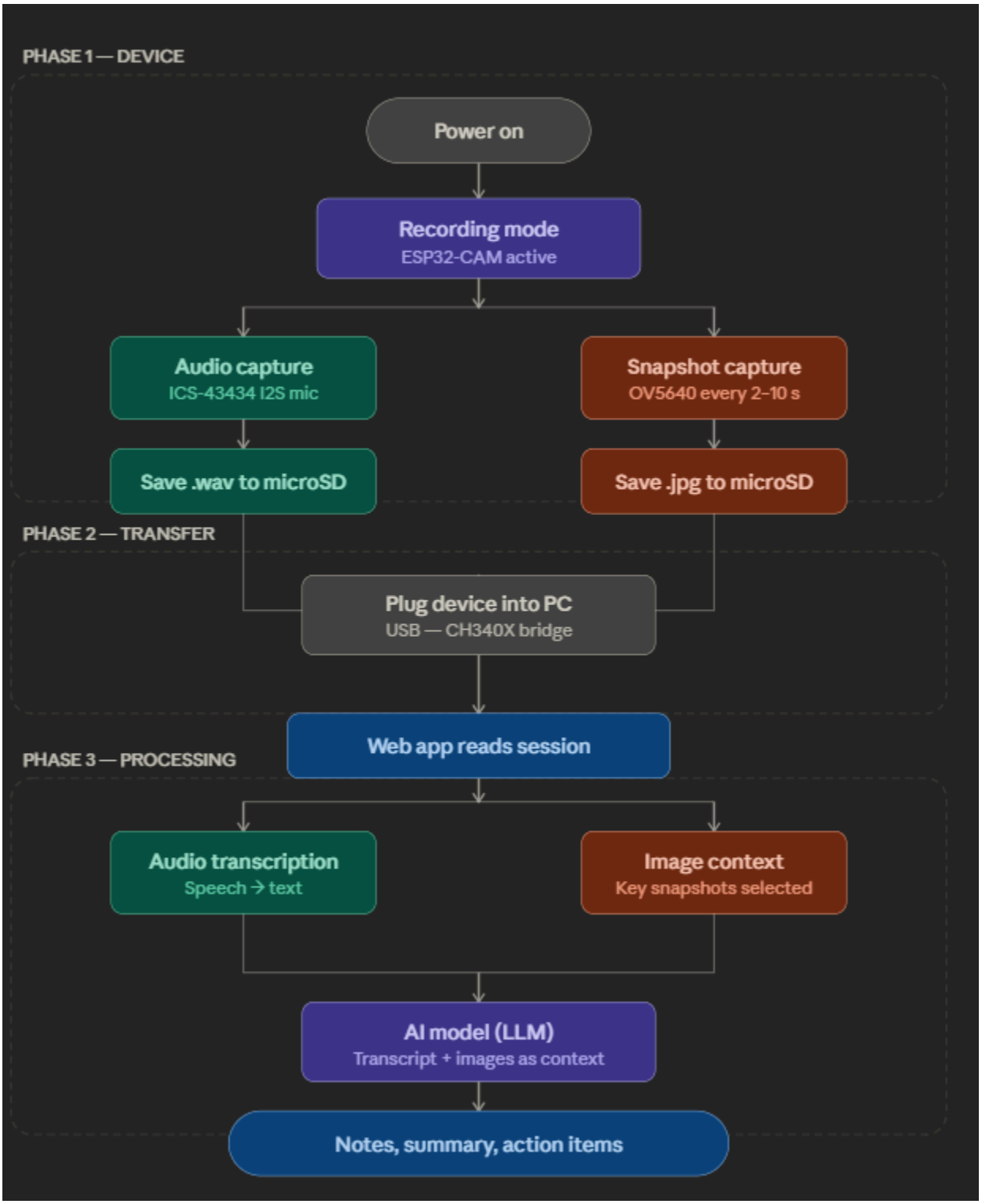
5.2 Web Application

- Device I/O: Web USB API + File System Access API
- Transcription: OpenAI Whisper API (audio → timestamped transcript)
- Note generation: multimodal LLM(Gemini 3, GPT-5, etc.) API (transcript + images → structured notes)
- Export: Download as Markdown, PDF, or copy to clipboard

6. Budget Estimate

Item	Qty	Est. Cost (USD)
ESP32-CAM (OV5640, 8MB PSRAM, CH340X)	1	
ICS-43434 I2S MEMS microphone breakout	1	~$5
microSD card 32 GB Class 10 (local)	1	
3.7 V LiPo 2000 mAh + JST-PH leads	1	~$7
Misc. wiring, headers, resistors	—	
3D-printed enclosure (filament)	—	~$3
Total (hardware)		**< $50 USD**

7. System Workflow



8. Future Scope

- Wi-Fi sync: Auto-upload sessions to a local NAS or cloud storage when on a known network.
- On-device wake word: Activate recording only when a specific phrase is detected (reduces storage).
- Speaker diarisation: Identify and label individual speakers in the transcript.
- Mobile companion app: View, search, and export notes directly from a smartphone.
- Multi-unit sync: Time-synchronise multiple SmartPuck devices in large lecture halls.

9. Conclusion

SmartPuck addresses a genuine gap in the recording-and-documentation market with a pragmatic, affordable hardware design and a modern AI-powered processing pipeline. The proposal is achievable with existing skills in ESP32 firmware development, React, and prompt engineering, and the sub-$50 hardware target makes the project viable for personal prototyping.

Upon successful completion, SmartPuck will stand as a functional proof-of-concept ready for further iteration, potential commercialisation, or submission as a university capstone project.

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